ICT 6101: Introduction to Information Security

- Introduction to information security: security principles, access control mechanisms, authentication schemes;
- Information security concepts, classification, security threats such as phishing, spam, adware, ransomware, trojan, virus, etc;
- Operating system security: classic security models, common vulnerabilities, Linux, Windows, and cloud security;
- Cyber-attacks: examples, tools, and methodologies,
- Network security: firewall, intrusion detection system; intrusion prevention system, cyber defense techniques;
- Cyber forensics: tools, mechanisms, challenges;
- Cyber-ethics: cybercrimes, intellectual properties, privacy.

What is cybersecurity?

Cyber security is the practice **of defending computers, servers, mobile devices, electronic systems, networks**, and data from malicious attacks.

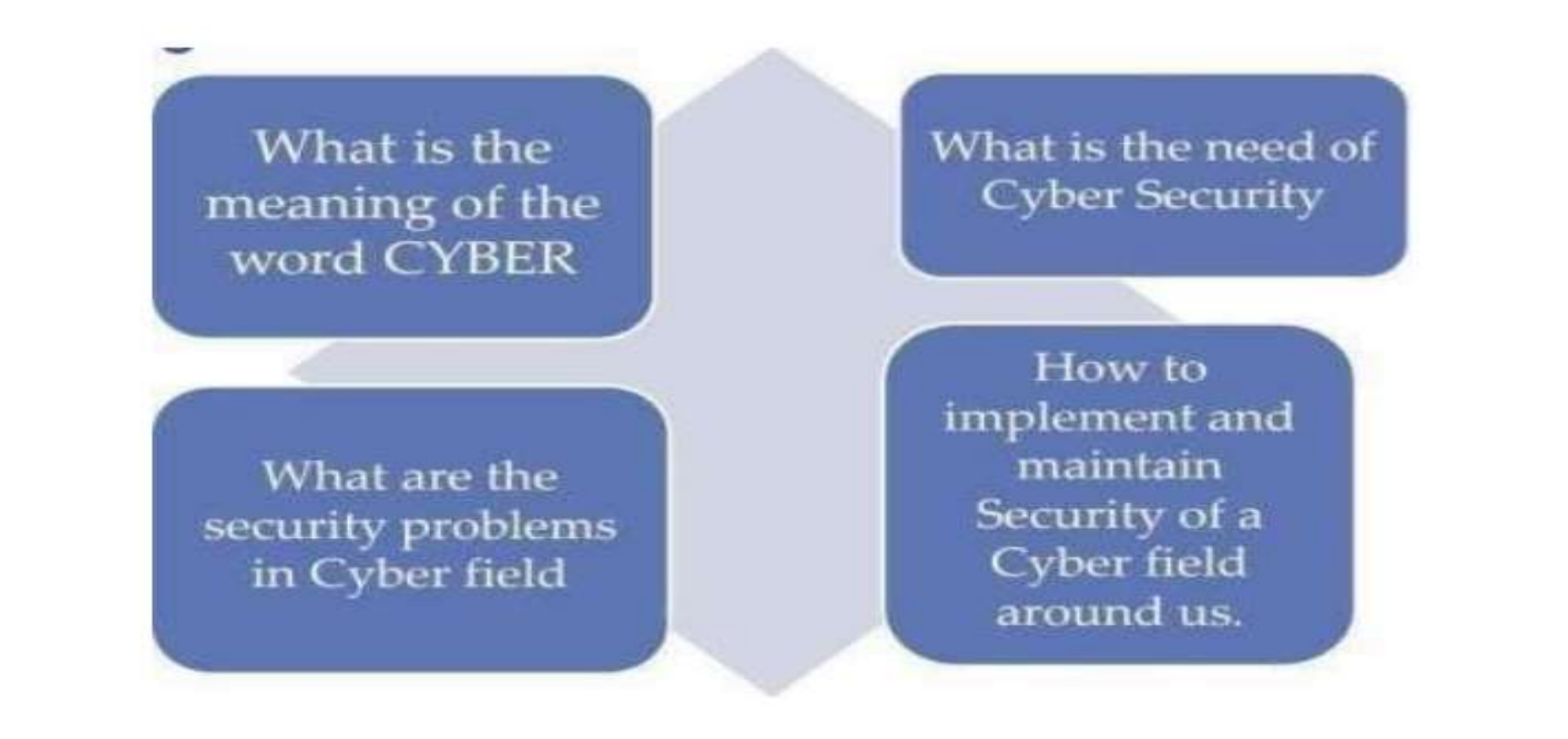
Or

Cyber security is **the application of technologies, processes, and controls to protect systems, networks, programs**, devices and data from cyber attacks.

Or

Cyber security refers **to every aspect of protecting an organization and its employees and assets against cyber threats**.

Understanding Cyber Security



Cyber and Cyberspace

- a combining form meaning “computer,” “computer network,” or “virtual reality,” - involving computers or computer networks (such as the Internet).

- Cyberspace can be defined as an intricate environment that involves interactions between people, software, and services. It is maintained by the worldwide distribution of information and communication technology devices and networks.

- the cyberspace today has become a common pool used by citizens, businesses, critical information infrastructure, military and governments in a fashion that makes it hard to induce clear boundaries among these different groups.

CIA triad

The three letters in " CIA triad " stand for Confidentiality, Integrity, and Availability. The CIA triad is a common model that forms the basis for the development of security systems. They are used for finding vulnerabilities and methods for creating solutions.



Confidentiality measures

Standard measures to establish confidentiality include:

- Data Encryption,
- Two-factor authentication,
- Biometric Verification,
- SecurityTokens



Integrity measures

Standard measures to guarantee Integrity include:

- Cryptography
- checksums Using file permissions
- Uninterrupted power supplies
- Data backups



Availability measures

Standard measures to guarantee Availability include:

- Backing up data to external drives,
- Implementing firewalls
- Having backup power supplies
- Data redundancy



Asset

- An asset is any data, device or other component of an organization's systems that is valuable – often because it contains sensitive data.
- For example:

- An employee's desktop computer, laptop or company phone would be considered an asset, as would applications on those devices.

- Likewise, critical infrastructure, such as servers and support systems, are assets.

- An organization's most common assets are information assets. These are things such as databases and physical files – i.e. the sensitive data that you store.

Threat/Cyber threat

- A threat is any incident that could negatively affect an asset – for example, if it's lost, knocked offline or accessed by an unauthorized party.

Or

- Cybersecurity threats are acts performed by individuals with harmful intent, whose goal is to steal data, cause damage to or disrupt computing systems.
- Threats can be categorized as circumstances that compromise the confidentiality, integrity or availability of an asset, and can either be intentional or accidental.
- Intentional threats include things such as criminal hacking or a malicious insider stealing information, whereas accidental threats generally involve employee error, a technical malfunction or an event that causes physical damage, such as a fire or natural disaster

Common Sources of Cyber Threats against organization

Here are several common sources of cyber threats against organizations:

- **Nation states**—hostile countries can launch cyber attacks against local companies and institutions, aiming to interfere with communications, cause disorder, and inflict damage.
- **Terrorist organizations**—terrorists conduct cyber attacks aimed at destroying or abusing critical infrastructure, threaten national security, disrupt economies, and cause bodily harm to citizens.
- **Criminal groups**—organized groups of hackers aim to break into computing systems for economic benefit. These groups use phishing, spam, spyware and malware for extortion, theft of private information, and online scams.
- **Hackers**—individual hackers target organizations using a variety of attack techniques. They are usually motivated by personal gain, revenge, financial gain, or political activity. Hackers often develop new threats, to advance their criminal ability and improve their personal standing in the hacker community.
- **Malicious insiders**—an employee who has legitimate access to company assets, and abuses their privileges to steal information or damage computing systems for economic or personal gain. Insiders may be employees, contractors, suppliers, or partners of the target organization. They can also be outsiders who have compromised a privileged account and are impersonating its owner.

Types of Cybersecurity Threats

- Malware Attacks
- Social Engineering Attacks
- Supply Chain Attacks
- Man-in-the-Middle Attack
- Denial-of-Service Attack
- Injection Attacks

Cyber attack

- An offensive action by a malicious actor that is intended to undermine the functions of networked computers and their related resources, including unauthorized access, unapproved changes, and malicious destruction.

- Examples of cyber attacks include Distributed Denial of Service (DDoS) and Man-in-the- Middle (MITM) attacks.

Threat, attack and risk differences

The terms cyber attack, cyber threat, and cyber risk are interrelated as follows:

- A **cyber attack** is an **offensive** action, whereas a **cyber threat i**s the **possibility** that a particular attack may occur, and the **cyber risk** associated with the subject threat **estimates the probability of potential losses** that may result.

- For example, a Distributed Denial of Service (**DDoS**) cyber attack by a **botnet** is a cyber threat for many enterprises with online retail websites, where the associated cyber risk is a function of **lost revenues due to website downtime** and the **probability that a DDoS cyber attack will occur**.

Cyber Security Solutions

Here are the main types of security solutions:

- **Application security**—used to test software application vulnerabilities during development and testing, and protect applications running in production, from threats like network attacks, exploits of software vulnerabilities, and web application attacks.
- **Network security—**monitors network traffic, identifies potentially malicious traffic, and enables organizations to block, filter or mitigate threats.
- **Cloud Security—**implements security controls in public, private and hybrid cloud environments, detecting and fixing false security configurations and vulnerabilities.
- **Endpoint security—**deployed on endpoint devices such as servers and employee workstations, which can prevent threats like malware, unauthorized access, and exploitation of operating system and browser vulnerabilities.
- **Internet of Things (IoT) security—**connected devices are often used to store sensitive data, but are usually not protected by design. IoT security solutions help gain visibility and improve security for IoT devices.
- **Threat intelligence**—combines multiple feeds containing data about attack signatures and threat actors, providing additional context for security events. Threat intelligence data can help security teams detect attacks, understand them, and design the most appropriate response.

What Are The Different Information Security Fields?

- Physical Cybersecurity **-** include biometric authentication, surveillance cameras, and access control systems to prevent unauthorized access.
- Information Security/endpoint security/application security
- IT Auditing & Consulting/Governance, risk, and compliance
- Ethical Hacking/Penetration Testing
- Intrusion detection
- Threat Intelligence
- Software Development
- Cloud security
- Digital Forensics and incident response
- Cryptography